

Introduction

Definition of Terms

1. Textiles

The word textile comes from the Latin word textilis "woven", which in turn comes from the Latin verb texere meaning "to weave". In textile science, however, a textile is freely defined as any product made from fibres; thus the term refers not only to woven fabrics but also to non woven fabrics knitted fabrics and special fabric constructions.

2. Fibre

Any substance, natural or man-made with a high length to width ratio and with suitable characteristics for being processed into fabric.

3. Yarn

An assemblage of fibres twisted or laid together so as to form a continuous strand that can be made into textile fabric.

4. Fabric

A plain substance constructed from solutions, fibres, yarns, fabrics, or any combinations of these.

5. Finish colour

Any process used to convert gray goods (unfinished fabric) into finished fabric.

Classification of Fibres

Classification means "Grouping together."

Textile fibres are classified according to;

(a) The Principle of origin.

Fibres either occur as fibrous forms in nature or they are manufactured, thus a major breakdown to indicate scientifically the origin of the fibre produces two major groups:

1. Natural group
2. Man-made fibre

(b) The general chemical type fibres can be protein, cellulosic, mineral, or synthesized also called non-cellulosic manmade.

(c) Generic term: grouping of like fibres of the manmade variety as provide by the textile fibre products Identification Act.

(d) Inclusion of common names of fibre and/or trade names many people are familiar with common names and trade names for fibres therefore examples of these can be included in any classification as a means of clarity the group name for the student and consumer.

Natural Fibres

Cellulosic Fibres

1. Seed hairs such as cotton, kapok
2. Bast fibres such as flax (linen) ramie, hemp, jute, kenaf
3. Leaf fibres such as pineapples abaca, agave e.g. sisal palm etc.

Protein fibres

1. Animal hairs e.g. wool (sheep), speciality hair fibres e.g.
2. Animal secretions e.g. silk

Mineral fibres

E.g. Asbestos (rock sources)

Rubber

E.g. Natural rubber

Man-Made Fibres

A. Man-made cellulosic fibres e.g.

1. Rayon – cuprammonium rayon (is no longer used in some countries, because it is carcinogenous-cancer causing).
2. Viscose- High-wet – modulus rayon.

B. Man-made modified cellulosic e.g.

1. Acetate
2. Triacetate

C. Man-made protein fibres e.g.

Azlon – not commonly used because it is very expensive to produce and the raw material (milk) is food.

D. Man-made non-cellulosic fibre e.g.

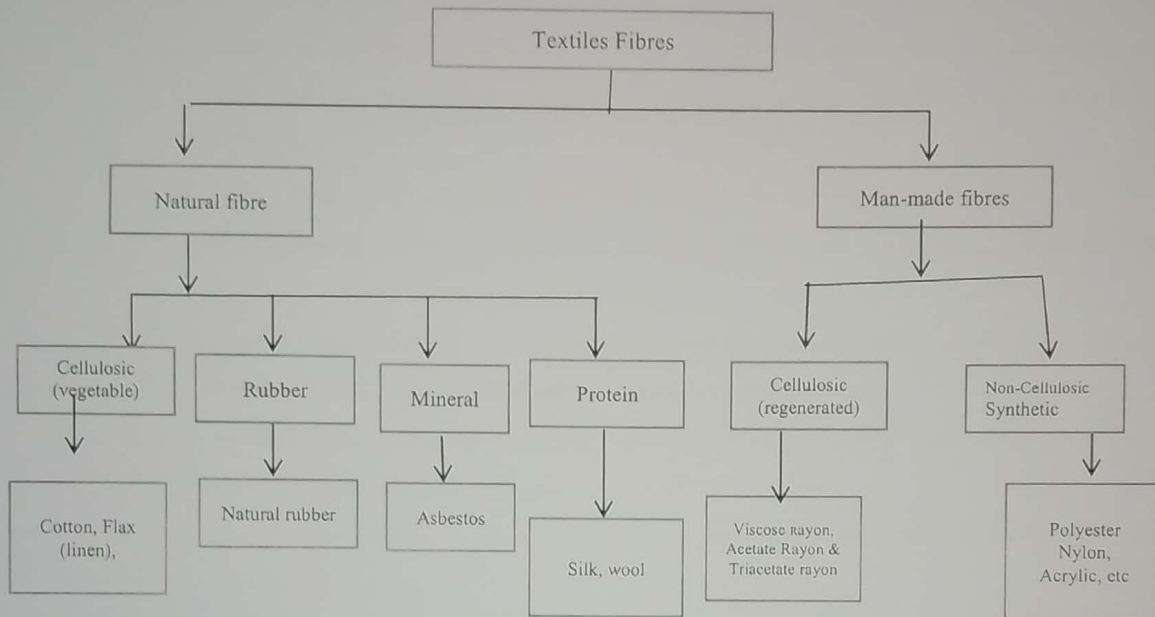
1. Polyamide fibres e.g. nylon and aramid
2. Polyester
3. Acrylic and

4. Modacrylic

E. Man-made mineral fibres e.g.

1. Glass
2. Metallic

Fibre Classification Table



Natural Cellulosic Fibres

Introduction

As explained earlier, natural cellulosic fibres are those that are found in nature but their main source is plant materials. They may be got from any part of the plant such as the stem, leaves, seed hairs or bark. In this lesson cotton and flax fibres will be discussed under the natural cellulosic fibers.

Cotton

Production

Cotton grows in any part of the world where the growing season is long and the climate is temperate to hot with adequate rainfall or irrigation. Cellulose will not form if the temperature is below 70° F.

Countries that produce cotton include Kenya, Egypt, India China America etc. Cotton grows on bushes 3 – 6 feet high.

The blossom appears, falls off and the ball begins its growth. Inside the ball are seeds from which fibre grows. When the ball is ripe it splits open and fluffy white fibres stand out like a powder puff. Cotton is picked by machine or hand. After picking the cotton is taken to a gin to remove the fibres from the seed. The fibres called lint are pressed into bales ready for spinning.

Processing of cotton

It involves several stages which include the following:

1. **Ginning** - The raw cotton is passed through a machine which separates seeds from the fibres and removes most the rest of the pieces of plant which may have become entangled.
2. **Baling** - this is the packing of cotton lint into bundles for shipping to factories.
3. **Bale Breaking** – At the spinning mills the bales are fed into machines which break down the matted fibres and then there of impurities.
4. **Opening and Cleaning** - it involves continuing the breaking down and cleaning process and producing an unbroken sheet of cotton fibre known as lap.
5. **Carding** – Separating the fibres and removing the short threads. This process produces a loose rope of cotton fibres known as “silver”.
6. **Combining** – Laying the fibres side by side by passing the slivers over revolving drums set with teeth.
7. **Drawing** – Pulling out on “drawing” the slivers and twisting several into one repeated times.
8. **Slubbing**– this is further drawing into a thin of sliver, now called a “roving” and winding the rovings into bobbins (this twists the threads slightly).
9. **Roving** – Drawing out a single thread fine enough for spinning.
10. **Spinning** – The further drawing out and twisting of the threads into a continuous length for weaving.
11. **Winding** – Removing faults in the thread and winding them into longer and more convenient reels.
12. **Doubling** – Twisting together two or more threads for greater strength.

Fibre Properties

Physical structure

Raw cotton is creamy in colour. The fibre is a single cell, which grows out of the seed as a cylindrical tube over a thousand times as long as it is thick. The quality of cotton depends on its length, strength, fineness and maturity.

Length: - Longer cotton fibres are finer and make stronger yarns.

Fineness: - Cotton fibre varies from 16 – 20 micrometers in diameter. The cross-sectional shape varies with maturity of the fibre.

Colour: - **Cotton** is naturally creamy white as it ages it becomes more beige, white, fibre is preferred.

Cotton Classification

This is the art of describing quality of cotton in terms of grade and staple length of the fiber.

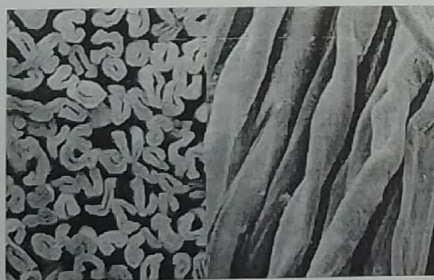
The staple length is identified by using sight and touch to determine the length of a representative bundle of fibres from a bale of cotton. The grade of cotton is determined by its appearance, more specifically its colour, leaf residue and preparation. Colour of cotton is described in terms of that range from white to yellow. It is also described in terms of lightness to darkness.

Microscopic properties

Cotton fibres are composed of an outer cuticle (skin) and a primary wall, a secondary wall and a central core of lumen. Immature fibres exhibit thin wall structures and a large lumen, whereas mature fibres have thick wall and small lumen that may not be continuous, because the wall closes the lumen in some sections. Fibres that have been swollen, as in mercerisation, do not show the twist as clearly as do untreated fibres and the cross-section tends to be smooth and round instead of flat and folded. Immature fibres may have an appearance similar to mercerized cotton.

The longitudinal view of regular cotton shows a ribbon like shape with twist (convolution) at regular intervals.

Cross-sectional viewlongitudinal view



Source: swicofil.com

Physical properties

1. Shape and appearance

Cotton fibres are fairly uniform in width, which varies between 12 and 20 microns, the central portion of the fibre is wider than either end, depending on variety and growing conditions, the length of the cotton fibre used in fabric manufacturing ranges from $\frac{1}{2}$ inch to $2\frac{1}{2}$ inches [1.27-6.35cm] category.

The colour fibres may vary from almost pure white in a dirty gray. Part of the colour may be due to impurities that can be removed during processing. High quality cotton fibre is usually very light cream or almost white.

2. Lusture

The lusture of cotton is low under finishes have been added such as mercerisation.

3. Tenacity strength

Cotton has moderate to above average strength when wet, cotton increases in strength, thus means that fibre care and what processing techniques do not require modification to compensate for reduced fibre strength when wet.

4. Resiliency

The resiliency of cotton is low.

5. Elastic recovery and elongation

Cotton has an elongation of 3 to 7 percent at the breaking point. It is relatively inelastic, with a recovery of only 75% at 2% extension.

6. Density

Cotton is one of the most dense of all commonly used fibres with a density of 1.54 to 1.56 grams per cubic centimetre.

7. Moisture regain

The moisture regain for cotton at standard conditions is 8.5%. Regain or absorbency at as percent relative humidity is approximately 15%, and at 100% relative humidity it may absorb 25 to 27% moisture. Wet cotton is not only stronger but also ore pliable than cotton.

Mercerized cotton fibres have been enlarged (swollen) and can absorb more moisture than non-mercerized fibres with standard regain of 8.5% - 10.3%.

8. Dimensional stability

Cotton fibres are relatively stable and do not stretch or shrink. Cotton fabrics, however do tend to shrink as a result of tensions encountered during yarn and fabric manufacture; consequently the fabrics require treatment to render them susceptible to shrinkage.

Thermal properties

Cotton burns readily and quickly, with a smell similar to burning paper. It leaves a small amount of fluffy gray ash. Normal exposure to heat encountered in routine care and processing will not damage cotton, but it will scorch if ironed at too high temperatures. Finishes such as starch, increase the tendency to scorch.

Chemical properties

1. Effects of alkalis

Cotton is highly resistant to alkalis. In fact, they are used in finishing and processing the fibre. Most detergents and laundry aids are alkalis, so cotton can be launched in these solutions without damage.

2. Effects of acids

Strong acids destroy cotton and hot dilute acids cause gradual fibre degradation, but the process is slow and may be immediately evident.

3. Effect of organic solvents

Cotton is highly resistant to most organic solvents and to all those used in normal care and stain removal. It is however, soluble in such compound as cupramonium hydroxide and supriethylene diamine; these solvent, therefore should be avoided except if used in the chemical analysis and identification of cotton.

4. Effects of sunlight, age and miscellaneous factors

Prolonged exposure to sunlight will cause the cotton fibre to become yellow and will gradually cause degradation. This damage is accelerated in the presence of moisture, some vat dyes, and some sulfur dyes. If properly stored, cotton will retain most of its strength and appearance over a long period. Cotton fabric should be stored in dry areas.

5. Static charges

Cotton does not build up static charge.

Biological properties

- (a) **Micro organism** - cotton is damaged by fungi such as mildew and by bacteria. Mildew produces a disagreeable odour and will result in rotting and degradation of cotton.
- (b) **Insects** – Moths and beetles will not attack or damage cotton. Silver fish will eat cotton cellulose especially if it's heavily starched.

Use and Care of Cotton

Cotton is for apparel fabrics, household or domestic goods for home furnishings, an industrial applications. The ability of cotton to accept colour and finishes, combined with its comfort makes it desirable choice for the consumer. The comfort factors especially its ability to absorb moisture and its softer makes it ideal for domestic items such as towels, sheets and pillow cases. Cotton fabrics have good wearing qualities, excellent laudability, good absorbency, good colour fastness when proper dye stuffs are used, good pliability or flexibility and good resistance. Care of cotton is not demanding.

Most cotton fabrics can be laundered dried with hoe laundering equipment.